Algebra II with Trigonometry Chapter 6.4

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file: 4533650_alg-2trig-h-chp-6-4-note.pdf

Here is a comprehensive overview and study guide based on the provided document.

1. Document Summary

This document, from an Algebra 2 Trig/Honors class, covers the graphing of four trigonometric functions: tangent, cotangent, secant, and cosecant. It outlines their fundamental periodic properties, reciprocal identities, and the locations of their vertical asymptotes. Additionally, it provides specific formulas to determine the period and the appropriate x-intervals for graphing one complete cycle of transformed versions of these functions, formatted as $y = a \cdot \text{trig}(k(x - b))$.

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2. Core Definitions, Theorems, and Formulas

Fundamental Periods:

- **Tangent and Cotangent:** The base period is π.
- $\tan(x + \pi) = \tan x$
- $\cot(x + \pi) = \cot x$
- **Secant and Cosecant:** The base period is 2π.
- $\sec(x + 2\pi) = \sec x$
- $\csc(x + 2\pi) = \csc x$

Reciprocal Identities:

- Cosecant: $\csc x = \frac{1}{\sin x}$

- Secant: $\sec x = \frac{1}{\cos x}$

Vertical Asymptotes (n is an integer):

- $y = \tan x$: $x = \frac{\pi}{2} + n\pi$
- $y = \cot x$: $x = n\pi$
- $y = \sec x$: $x = \frac{\pi}{2} + n\pi$
- $y = \csc x$: $x = n\pi$

Transformation Formulas ($k > 0$): For functions in the form $y = a \cdot \text{trig}(k(x - b))$:

- **Period:**
- Tangent/Cotangent: $\frac{\pi}{k}$
- Secant/Cosecant: $\frac{2\pi}{k}$
- **Intervals for One Period:**
- $y = a\tan(k(x-b))$: $\left(-\frac{\pi}{2k} + b, \frac{\pi}{2k} + b\right)$
- $y = a\cot(k(x-b))$: $\left(0 + b, \frac{\pi}{k} + b\right)$
- $y = a\sec(k(x-b))$ or $y = a\csc(k(x-b))$: $\left(0 + b, \frac{2\pi}{k} + b\right)$

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3. Worked Study Guide: Step-by-Step Graphing Prep

To graph complex trigonometric transformations, you need to find the period and the starting/ending points of one standard cycle. Follow these steps:

Step 1: Factor the inside of the function. The formulas provided in the text require the function to be in the form $k(x - b)$. If you are given something like $(cx - d)$, you must factor out the c to find your k and b values: $c(x - \frac{d}{c})$.

- *Example:* $\sec(3x - \frac{\pi}{2})$ becomes $\sec(3(x - \frac{\pi}{6}))$. Here, $k = 3$ and $b = \frac{\pi}{6}$.

Step 2: Determine the new Period. Look at the base function.

- If it is $\tan$ or $\cot$, the period is $\frac{\pi}{k}$.
- If it is $\sec$ or $\csc$, the period is $\frac{2\pi}{k}$.

Step 3: Calculate the graphing interval. Use the specific interval formula for your function from Section 2. Plug in your k and b values.

- *For Tangent:* You are shifting the standard interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ by adjusting for the horizontal stretch (k) and the phase shift (b).
- *For Cot, Sec, Csc:* You are shifting the standard starting point of 0 to your phase shift b, and ending at the end of your new period plus the phase shift.

Step 4: Use the interval to frame your graph. The x-values you calculated in Step 3 dictate the left and right boundaries of one full wave. For tangent and cotangent, these boundaries perfectly align with your vertical asymptotes.

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4. Practice Problems

(Note: Problems 1-4 are taken directly from the blank exercises on page 3 of the document. Problem 5 is derived from the core concepts).

Problem 1: Find the period and the appropriate graphing interval for one period of $y = 4\tan(4x - 2\pi)$. **Answer:**

- Factor: $y = 4\tan(4(x - \frac{\pi}{2}))$. Therefore, $k = 4, b = \frac{\pi}{2}$.
- Period: $\frac{\pi}{k} = \mathbf{\frac{\pi}{4}}$
- Interval: $(-\frac{\pi}{2(4)} + \frac{\pi}{2}, \frac{\pi}{2(4)} + \frac{\pi}{2}) = (-\frac{\pi}{8} + \frac{4\pi}{8}, \frac{\pi}{8} + \frac{4\pi}{8}) = \mathbf{(\frac{3\pi}{8}, \frac{5\pi}{8})}$

Problem 2: Find the period and the appropriate graphing interval for one period of $y = 2\csc(\pi x - \frac{\pi}{3})$. **Answer:**

- Factor: $y = 2\csc(\pi(x - \frac{1}{3}))$. Therefore, $k = \pi, b = \frac{1}{3}$.
- Period: $\frac{2\pi}{k} = \frac{2\pi}{\pi} = \mathbf{2}$
- Interval: $(0 + \frac{1}{3}, \frac{2\pi}{\pi} + \frac{1}{3}) = (\frac{1}{3}, 2 + \frac{1}{3}) = \mathbf{(\frac{1}{3}, \frac{7}{3})}$

Problem 3: Find the period and the appropriate graphing interval for one period of $y = 5\sec(3x - \frac{\pi}{2})$. **Answer:**

- Factor: $y = 5\sec(3(x - \frac{\pi}{6}))$. Therefore, $k = 3, b = \frac{\pi}{6}$.
- Period: $\frac{2\pi}{k} = \mathbf{\frac{2\pi}{3}}$

- Interval: $(0 + \frac{\pi}{6}, \frac{2\pi}{3} + \frac{\pi}{6}) = (\frac{\pi}{6}, \frac{4\pi}{6} + \frac{\pi}{6}) = \boldsymbol{(\frac{\pi}{6}, \frac{5\pi}{6})}$

Problem 4: Find the period and the appropriate graphing interval for one period of $y = 2\cot(3\pi x + 3\pi)$. **Answer:**

- Factor: $y = 2\cot(3\pi(x + 1)) = 2\cot(3\pi(x - (-1)))$. Therefore, $k = 3\pi, b = -1$.
- Period: $\frac{\pi}{k} = \frac{\pi}{3\pi} = \boldsymbol{\frac{1}{3}}$
- Interval: $(0 + (-1), \frac{\pi}{3\pi} + (-1)) = (-1, \frac{1}{3} - 1) = \boldsymbol{(-1, -\frac{2}{3})}$

Problem 5: Based on the document, what is the identity used to graph $y = \csc x$, and where do its vertical asymptotes occur? **Answer:** The identity used is $\csc x = \frac{1}{\sin x}$. The vertical asymptotes occur wherever sine is zero, which is at $x = n\pi$ (where n is an integer).

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5. Assumptions and Ambiguities Resolved

- **Factoring for Transformations:** The document gives the formula for transformations in the form $k(x - b)$. However, the examples on page 3 are formatted as $(cx - d)$. The document assumes the student already knows how to algebraically factor c out to reveal the true horizontal stretch (k) and phase shift (b). I made this step explicit in the study guide and practice problems.
- **Bracket vs. Parenthesis Notation:** In Section II (Page 2), the document states the interval for $y = a\csc(kx)$ is $[0, \frac{2\pi}{k}]$ (using brackets, meaning inclusive). However, immediately below it, the shifted interval formula uses parentheses: $(0 + b, \frac{2\pi}{k} + b)$. Because cosecant is undefined at 0 and π, using parentheses to denote the boundaries between asymptotes is technically more mathematically precise. I chose to rely on the parenthesis formulation for the practice problems as it perfectly aligns with the phase-shift formula provided in the text.
- **Blank Example Page:** Page 3 contains problems but no solutions. I resolved this by solving those exact problems to fulfill the requirement for five practice problems based directly on the text.